

Text as numbers: tokens and embeddings

A network multiplies matrices; a sentence is not a matrix. Two steps close the gap. Tokenisation cuts text into pieces and numbers them. Embeddings turn each number into a point in space, placed so that words used alike sit close together — and meaning becomes something you can measure with a dot product.

Unit 5 · Language

80 minutes

Live, then explored · no lab

Vectors loaded, not trained

Tokens

One-hot

Similarity

Loaded, not trained

Explore

Before Tuesday

PART 1 · 15 MINUTES

Tokens and a vocabulary

1

The simplest tokeniser splits on spaces and gives every distinct word an ID. “*Grand-Bassam students study artificial intelligence*” becomes five tokens and five integers. It breaks immediately on real text: a word never seen in training has no ID, and *study*, *studies* and *studying* become three unrelated entries.

SUBWORD TOKENISATION

Modern language models learn a vocabulary of frequent **pieces** — whole common words, plus fragments for everything else. A rare word splits into known parts, so nothing is ever out of vocabulary. GPT-2's tokeniser has 50,257 pieces. Names, numbers and many African-language words split into more pieces than common English words do — which is part of why models cost more, and often perform worse, on them.



This connects to session 3

The gap between well-served and poorly served languages you saw on the leaderboards starts here, at the tokeniser: text that splits into many small pieces uses more of the context window and gives the model less to hold on to per piece.

PART 2 · 15 MINUTES

Why an ID alone is not enough

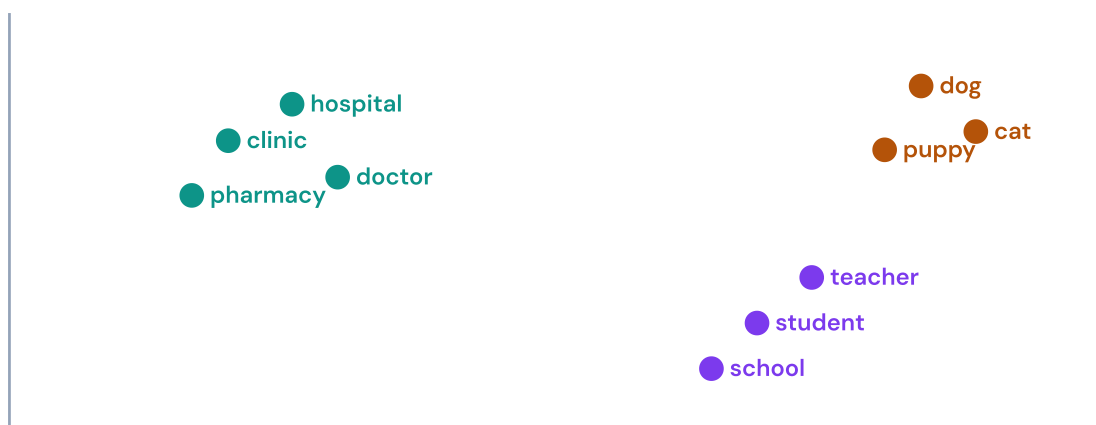
2

Token 3 is not "more" than token 2, so an ID cannot go into a network as a number. The honest first encoding is **one-hot**: a vector as long as the vocabulary, all zeros except a 1 at the token's position. It works, and it has two fatal properties: with 50,000 tokens every word is a 50,000-long vector, and every pair of different words is exactly equally unrelated — *clinic* is as far from *hospital* as from *elephant*.

AN EMBEDDING

Give every token a short, dense vector of learned numbers — 50 or a few hundred instead of 50,000 — and place them so that **tokens used in similar contexts end up close together**. An embedding table is a weight matrix of shape (vocabulary, dimensions); looking up a token is selecting its row.

NEARNESS IN THE SPACE STANDS IN FOR NEARNESS IN USE



a 2-D sketch of a space that really has 50 or more dimensions

Measuring meaning: cosine similarity

Run at the front from CSC4315_S24_Explore_Tokens_And_Embeddings.

3

COSINE SIMILARITY

$$\cos(a, b) = a \cdot b / (\|a\| \|b\|)$$

The dot product of two vectors divided by their lengths: 1 when they point the same way, 0 at right angles, -1 when opposite. It ignores length and compares direction.

By hand, for $a = [1, 2]$ and $b = [2, 3]$: $a \cdot b = 8$, $\|a\| = \sqrt{5} \approx 2.236$, $\|b\| = \sqrt{13} \approx 3.606$, so $\cos = 8 / 8.062 \approx \mathbf{0.992}$ — nearly the same direction.

from the offline teaching vectors

TEXT

nearest to clinic: clinic 1.0 · pharmacy 0.95 · hospital 0.924 · beach 0.784 · grand-bassam 0.703
king - man + woman → queen 0.951 · princess 0.933 · girl 0.769 · woman 0.561



Which vectors the room is looking at

With internet access the notebook loads real 50-dimensional GloVe vectors, trained on Wikipedia and news text — 400,000 words. Offline, it uses 34 **synthetic teaching vectors** built by hand on eight readable axes. They make the mechanics runnable and look far tidier than real embeddings. Real ones are noisier, and more interesting.

The famous analogy, read carefully

king - man + woman lands near *queen*. It is a striking result and a fragile one. Reported versions usually exclude the three input words from the answers, because the nearest vector is often one of them. Many analogies simply fail. And the geometry reflects whatever was in the training text — including stereotypes: real embeddings have placed occupations closer to one gender than the other because the text did.

SEPARATE THE GEOMETRY FROM THE STORY

For every result, write two things apart: **what the geometry returned** — a list of words and numbers — and **your interpretation** of why. The vectors do not understand *royalty*; they encode which words appeared near which other words.

PART 4 · 10 MINUTES

Why loaded, not trained

4

Embeddings that behave well are learned by predicting words from their neighbours over billions of words of text — far more data and time than one session has. So we load them, exactly as session 23 loaded a pretrained image network. In a language model the embedding table is simply the first layer, trained with everything else.

Three questions

Q1. An embedding table has 50,257 rows and 768 columns. What is a row, and what is a column? A row is one token's vector; a column is one learned dimension, shared by every token.

Q2. Why cosine rather than plain distance? Frequent words tend to have longer vectors. Cosine compares direction only, so frequency does not masquerade as meaning.

Q3. The nearest neighbour of *nurse* is a gendered word in real embeddings. Is that a fact about nurses? No — it is a fact about the text the vectors were trained on. The model inherited it, and a system built on it would repeat it unless someone checks.

EXPLORE · 20 MINUTES · NO LAB TO HAND IN

Walk around the space

5

Open `CSC4315_S24_Explore_Tokens_And_Embeddings`. Neighbours of three words, the analogy, and three similarities to put in order before computing. Nothing to submit; keep `nearest`.

Before Tuesday

1. **Without notes:** tokenisation, why subwords, what an embedding is, and the cosine formula.
2. **Bring a calculator** — Tuesday you compute attention by hand.



Session 25 — Tuesday 01 December, 16:00

Attention, computed by hand: queries, keys and values on a three-token sentence, with small integers.